Advanced Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Compound angle formulas**

Expand $\sin(x + 30^\circ)$ in terms of $\sin x$ and $\cos x$, using exact values for $\sin 30^\circ$ and $\cos 30^\circ$.

$\sin(x + 30^\circ) = \sin x \cos 30^\circ + \cos x \sin 30^\circ = \frac{\sqrt{3}}{2}\sin x + \frac{1}{2}\cos x$.

2 **[Foundational]** • **Target: Compound angle formulas**

Given $\sin A = \frac{3}{5}$ and $\cos B = \frac{12}{13}$, with A and B both acute, find $\cos(A + B)$.

Since A, B are acute, $\cos A = \frac{4}{5}$ and $\sin B = \frac{5}{13}$. $\cos(A + B) = \cos A \cos B - \sin A \sin B = \frac{4}{5} \cdot \frac{12}{13} - \frac{3}{5} \cdot \frac{5}{13} = \frac{48}{65} - \frac{15}{65} = \frac{33}{65}$.

3 **[Foundational]** • **Target: Compound angle formulas**

Use the compound-angle formula for tangent to find the exact value of $\tan 75^\circ$, using $75^\circ = 45^\circ + 30^\circ$.

$\tan 75^\circ = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ} = \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$. Rationalizing: $\frac{(\sqrt{3} + 1)^2}{(\sqrt{3} - 1)(\sqrt{3} + 1)} = \frac{4 + 2\sqrt{3}}{2} = 2 + \sqrt{3}$.

4 **[Foundational]** • **Target: Compound angle formulas**

Use the compound-angle formula for cosine to find the exact value of $\cos 15^\circ$, using $15^\circ = 45^\circ - 30^\circ$.

$\cos 15^\circ = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}$.

5 **[Foundational]** • **Target: Double angle formulas**

Given $\sin\theta = \frac{4}{5}$ with θ acute, find $\sin 2\theta$ and $\cos 2\theta$.

Since θ is acute, $\cos\theta = \frac{3}{5}$. $\sin 2\theta = 2\sin\theta\cos\theta = 2\left(\frac{4}{5}\right)\left(\frac{3}{5}\right) = \frac{24}{25}$.
$\cos 2\theta = 1 - 2\sin^2\theta = 1 - 2\left(\frac{16}{25}\right) = -\frac{7}{25}$.

6 **[Foundational]** • **Target: Double angle formulas**

Given $\cos\theta = -\frac{1}{3}$ with $90^\circ < \theta < 180^\circ$, find $\cos 2\theta$ and state which quadrant 2θ lies in.

$\cos 2\theta = 2\cos^2\theta - 1 = 2\left(\frac{1}{9}\right) - 1 = -\frac{7}{9}$. Since $90^\circ < \theta < 180^\circ$, 2θ lies between 180° and 360°; combined with $\cos 2\theta < 0$ (which requires an angle between 90° and 270°), 2θ must lie in the third quadrant, i.e. $180^\circ < 2\theta < 270^\circ$.

7 **[Foundational]** • **Target: Double angle formulas**

Given $\tan\theta = \frac{1}{2}$ with θ acute, find $\tan 2\theta$, using $\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$.

$\tan 2\theta = \frac{2\left(\frac{1}{2}\right)}{1 - \left(\frac{1}{2}\right)^2} = \frac{1}{1 - \frac{1}{4}} = \frac{1}{3/4} = \frac{4}{3}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: The R sin(θ+α) form**

Express $3\sin\theta + 4\cos\theta$ in the form $R\sin(\theta + \alpha)$, with $R > 0$ and $0^\circ < \alpha < 90^\circ$, giving α to the nearest degree.

$R = \sqrt{3^2 + 4^2} = 5$. $\tan\alpha = \frac{4}{3} \Rightarrow \alpha \approx 53^\circ$. So $3\sin\theta + 4\cos\theta = 5\sin(\theta + 53^\circ)$.

9 **[Intermediate]** • **Target: The R sin(θ+α) form**

Express $\sqrt{3}\sin\theta - \cos\theta$ in the form $R\sin(\theta - \alpha)$, giving exact values of R and α.

$R = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{4} = 2$. $\tan\alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = 30^\circ$. So $\sqrt{3}\sin\theta - \cos\theta = 2\sin(\theta - 30^\circ)$.

10 **[Intermediate]** • **Target: Using the R-form to find extrema and solve equations**

The graph shows $y = 5\sin(x + 53^\circ)$, the R-form of $3\sin x + 4\cos x$ found earlier. State the maximum and minimum values of $3\sin\theta + 4\cos\theta$, and the value of θ (to the nearest degree, $0^\circ \le \theta < 360^\circ$) at which the maximum occurs.

Since $-1 \le \sin(\theta + 53^\circ) \le 1$, the maximum value is $R = 5$ and the minimum is -5. The maximum occurs when $\theta + 53^\circ = 90^\circ$, i.e. $\theta = 37^\circ$.

11 **[Intermediate]** • **Target: Using the R-form to find extrema and solve equations**

Solve $\sqrt{3}\sin\theta - \cos\theta = 1$ for $0^\circ \le \theta < 360^\circ$, using the R-form $2\sin(\theta - 30^\circ)$ found earlier.

$2\sin(\theta - 30^\circ) = 1 \Rightarrow \sin(\theta - 30^\circ) = \frac{1}{2}$. For $\theta - 30^\circ$ in the range $-30^\circ \le \theta - 30^\circ < 330^\circ$: $\theta - 30^\circ = 30^\circ$ or 150°, giving $\theta = 60^\circ$ or $\theta = 180^\circ$.

12 **[Intermediate]** • **Target: Solving equations that combine several identities**

Solve $2\cos 2\theta + 3\sin\theta = 1$ for $0^\circ \le \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$, giving answers to one decimal place where needed.

$2(1 - 2\sin^2\theta) + 3\sin\theta = 1 \Rightarrow 2 - 4\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 4\sin^2\theta - 3\sin\theta - 1 = 0 \Rightarrow (4\sin\theta + 1)(\sin\theta - 1) = 0$. So $\sin\theta = 1$, giving $\theta = 90^\circ$; or $\sin\theta = -\frac{1}{4}$, giving $\theta \approx 194.5^\circ$ or $\theta \approx 345.5^\circ$.

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13 [Intermediate] • Target: A well-designed word problem: alternating current

This question has three parts. The voltage in an AC circuit is modeled by $V(t) = 6\sin(100\pi t) + 8\cos(100\pi t)$ volts, where t is time in seconds. (a) Express $V(t)$ in the form $R\sin(100\pi t + \alpha)$, giving R exactly and α in radians to four decimal places. (b) State the maximum voltage reached, and the smallest positive value of t (in seconds, to four significant figures) at which it occurs. (c) Find the two values of t in $0 \le t < 0.02$ at which $V(t) = 5$, to four significant figures.

(a) $R = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$. $\alpha = \arctan\left(\frac{8}{6}\right) \approx 0.9273$ rad. So $V(t) = 10\sin(100\pi t + 0.9273)$. (b) The maximum voltage is $R = 10$ volts, occurring when $100\pi t + 0.9273 = \frac{\pi}{2}$, i.e. $t = \frac{1.5708 - 0.9273}{100\pi} \approx 0.002049$ s. (c) Solve $10\sin(100\pi t + 0.9273) = 5 \Rightarrow \sin(100\pi t + 0.9273) = 0.5$. Using $100\pi t + 0.9273 = \frac{5\pi}{6} \approx 2.6180$: $t = \frac{2.6180 - 0.9273}{100\pi} \approx 0.005382$ s. Using $100\pi t + 0.9273 = \frac{\pi}{6} + 2\pi \approx 6.8068$.

14 [Intermediate] • Target: Proving trigonometric identities

Level 3 · Advanced A tank initially contains 100 liters of pure water. Brine containing 0.5 kg of salt per liter enters the tank at a rate of 4 L/min, and the well-stirred solution leaves the tank at the same rate of 4 L/min. Find the amount of salt $S(t)$ in the tank after t minutes.

Rate of change of salt: $\frac{dS}{dt} = \text{Rate In} - \text{Rate Out}$. Rate In $= (0.5 \text{ kg/L}) \times (4 \text{ L/min}) = 2 \text{ kg/min}$. Since volume remains constant at 100 L, salt concentration is $S(t)/100$.

Rate Out $= \left(\frac{S(t)}{100} \text{ kg/L}\right) \times (4 \text{ L/min}) = \frac{4S}{100} = 0.04S$ kg/min. Set up the linear ODE:

$\frac{dS}{dt} = 2 - 0.04S = -0.04(S - 50)$ Separate variables.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solving equations that combine several identities • Cross-Lesson: 4.01 Differentiation

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

Substitute $\cos(2\theta) = 1 - 2\sin^2\theta$ to express the entire equation in terms of $\sin\theta$: $(1 - 2\sin^2\theta) + 3\sin\theta = 2$. $-2\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0$. Factor the quadratic in $\sin\theta$: $(2\sin\theta - 1)(\sin\theta - 1) = 0$. Case 1: $2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{1}{2}$. On $[0, 2\pi)$, $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$. Case 2: $\sin\theta - 1 = 0 \Rightarrow \sin\theta = 1$. On.

16 [Challenging] • Target: A well-designed word problem: alternating current • Cross-Lesson: 5.01 Integration

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.

Expand the double-angle expressions on the left-hand side: LHS $= \frac{2\sin x\cos x}{\sin x} - \frac{2\cos^2 x - 1}{\cos x}$. Cancel $\sin x$ in the first term and combine over a common denominator of $\cos x$: $= 2\cos x - \frac{2\cos^2 x - 1}{\cos x} = \frac{2\cos^2 x - (2\cos^2 x - 1)}{\cos x}$. $= \frac{2\cos^2 x - 2\cos^2 x + 1}{\cos x} = \frac{1}{\cos x} = \sec x = \text{RHS}$. ✓

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17 **[Challenging]** • **Target: Compound angle formulas** • **Cross-Lesson: 6.01 Sequences & Series**

Express $z = -1 + i\sqrt{3}$ in polar form $r(\cos\theta + i\sin\theta)$ and exponential form $re^{i\theta}$, with principal argument $-\pi < \theta \leq \pi$. Hence find z^6 and determine all three solutions to $w^3 = z$.

1. Modulus: $r = |z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2$. Argument: since z lies in Quadrant II, $\theta = \pi - \arctan(\sqrt{3}) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$. Polar form: $z = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$; Exponential form: $z = 2e^{i2\pi/3}$. 2. Compute z^6 via De Moivre's Theorem: $z^6 = 2^6 e^{i\cdot 6(2\pi/3)} = 64e^{i4\pi} = 64(\cos 4\pi + i\sin 4\pi) = 64(1+0i) = 64.$ 3. Solve $w^3 = z = 2e^{i2\pi/3}$.

18 **[Challenging]** • **Target: Double angle formulas** • **Cross-Lesson: 7.01 Vectors**

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.

Step 1: Identify smallest angle: The smallest angle is A, opposite the shortest side $a = 7$. Step 2: Law of Cosines for angle A: $\cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{9^2+10^2-7^2}{2(9)(10)} = \frac{81+100-49}{180} = \frac{132}{180} = \frac{11}{15} \approx 0.7333.$ $A = \arccos(11/15) \approx 42.8^\circ$. Step 3: Heron's formula for area: Semi-perimeter $s = \frac{7+9+10}{2} = 13.$.

19 **[Challenging]** • **Target: Solving equations using double angle formulas** • **Cross-Lesson: 8.01 Algebra: Partial Fractions & Division**

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

Step 1: Determine the rotation angle θ: Here $A = 5$, $B = -4$, and $C = 8$. $\cot(2\theta) = \frac{A-C}{B} = \frac{5-8}{-4} = \frac{-3}{-4} = \frac{3}{4} > 0.$ Since $\cot(2\theta) = 3/4$, by right triangle trigonometry $\cos(2\theta) = \frac{3}{5}$. Using half-angle formulas for $\theta \in (0, \pi/4)$: $\cos\theta = \sqrt{\frac{1+\cos(2\theta)}{2}} = \sqrt{\frac{1+3/5}{2}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}},.$

20 **[Challenging]** • **Target: The R sin(θ+α) form** • **Cross-Lesson: 9.01 Complex Numbers**

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

Step 1: Express -16 in polar modulus-argument form: $z^4 = -16 = 16(\cos\pi + i\sin\pi).$ Here $r = 16$, $\theta = \pi$, and $n = 4$. Step 2: Apply n-th roots formula: $w_k = \sqrt[4]{16}\left[\cos\left(\frac{\pi+2\pi k}{4}\right) + i\sin\left(\frac{\pi+2\pi k}{4}\right)\right] = 2\left[\cos\left(\frac{(2k+1)\pi}{4}\right) + i\sin\left(\frac{(2k+1)\pi}{4}\right)\right],\ k \in \{0,1,2,3\}.$ Step 3: Evaluate for each k:$k = 0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Compound angle formulas

Express $5\sin\theta + 12\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 7

b. **13** ✓

c. 17

d. 14

$R = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$.

22 [Foundational] • Target: Double angle formulas

Solve $\sin(\theta) = \frac{\sqrt{2}}{2}$ for $\theta \in [0^\circ, 360^\circ)$. How many solutions are there?

a. **2** ✓

b. 3

c. 1

d. 4

Solutions: $45^\circ, 135^\circ$. Count: 2.

23 [Foundational] • Target: Solving equations using double angle formulas

Find the maximum value of $8\sin\theta + 15\cos\theta$ as θ varies over all real values.

a. **17** ✓

b. 23

c. 15

d. 8

Maximum value $= R = \sqrt{8^2 + 15^2} = 17$.

24 [Foundational] • Target: The R sin(θ+α) form

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1}{4}$

b. $\frac{1}{2}$ ✓

c. $\frac{3}{5}$

d. $-\frac{1}{2}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{1}{2}\right)}{2}} = \frac{1}{2}$.

25 [Foundational] • Target: Using the R-form to find extrema and solve equations

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 10)

a. $6 - 24$

b. 22

c. $2 - 64$

d. $6 + 24$ ✓

$(45^+ 30^) = 45^3 0^+ 45^3 0^= 6 + 24$.

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26 [Foundational] • Target: Proving trigonometric identities

Find the exact value of $\cos 75^\circ$.

a. $\frac{\sqrt{6}-\sqrt{2}}{4}$ ✓

b. $\frac{-\sqrt{2}+\sqrt{6}}{4}$

c. $\frac{\sqrt{6}+\sqrt{2}}{4}$

d. $\frac{-\sqrt{6}+\sqrt{2}}{4}$

$\cos 75^\circ = \frac{\sqrt{6}-\sqrt{2}}{4}$.

27 [Foundational] • Target: Solving equations that combine several identities

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$ ✓

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

We are given $\cos\alpha = \frac{3}{5}$ and α is in the first quadrant. Using the identity $\sin^2\alpha + \cos^2\alpha = 1$, we find $\sin\alpha = \sqrt{1-(\frac{3}{5})^2} = \sqrt{1-\frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$ (since α is in the first quadrant, $\sin\alpha$ is positive). We are given $\sin\beta = -\frac{4}{5}$ and β is in the third quadrant. Using the identity $\sin^2\beta + \cos^2\beta = 1$, we find.

28 [Foundational] • Target: A well-designed word problem: alternating current

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

The problem states that point $P(x, y)$ lies on the unit circle, meaning $x^2 + y^2 = 1$. We are given that the y-coordinate of P is $y = \frac{3}{5}$. Substituting this into the unit circle equation, we get $x^2 + (\frac{3}{5})^2 = 1$. This simplifies to $x^2 + \frac{9}{25} = 1$. Subtracting $\frac{9}{25}$ from both sides gives $x^2 = 1 - \frac{9}{25} = \frac{16}{25}$. Taking the square root of both sides, we get $x = \pm\frac{4}{5}$. The problem also states that point P is in the second quadrant, where the x.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Compound angle formulas

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$ ✓

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

For any point on the unit circle, the coordinates are $(\cos(\theta), \sin(\theta))$. The equation of the unit circle is $x^2 + y^2 = 1$. Substituting the coordinates, we get $\cos^2(\theta) + \sin^2(\theta) = 1$. We are given that $\cos(\theta) = \frac{1}{2}$. So, $\left(\frac{1}{2}\right)^2 + \sin^2(\theta) = 1$. This simplifies to $\frac{1}{4} + \sin^2(\theta) = 1$. Subtracting $\frac{1}{4}$ from both sides gives $\sin^2(\theta) = 1 - \frac{1}{4} = \frac{3}{4}$. Taking the square root of both sides, we get.

30 [Intermediate] • Target: Double angle formulas

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. **1** ✓

d. 5/3

The point P is on the unit circle, so its coordinates are $(\cos\theta, \sin\theta)$. We are given $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. The point Q has coordinates (x, y) and its distance from the origin is 5. The angle between the positive x-axis and OQ is α. Thus, $x = 5\cos\alpha$ and $y = 5\sin\alpha$. We are asked to find the maximum possible value of the product of the x-coordinates of P and Q, which is $\cos\theta \cdot x$. Substituting the given values, we h.

31 [Intermediate] • Target: Solving equations using double angle formulas

Using the compound angle formula, expand $\sin(x + 90^\circ)$.

a. $\cos x$ ✓

b. $-\cos x$

c. $\sin x$

d. $-\sin x$

$\sin(x + 90^\circ) = \sin x \cos 90^\circ + \cos x \sin 90^\circ = 0 + \cos x = \cos x$.

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32 [Intermediate] • Target: The R sin(θ+α) form

If $\cos\theta = \frac{4}{5}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{4}{5}$

b. $-\frac{3}{5}$

c. $\frac{3}{5}$ ✓

d. $\frac{7}{10}$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{16}{25}} = \frac{3}{5}$.

33 [Intermediate] • Target: Using the R-form to find extrema and solve equations

If $\sin\theta = \frac{3}{5}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{37}{50}$

b. $\frac{16}{25}$ ✓

c. $\frac{4}{5}$

d. $\frac{9}{25}$

$\cos^2\theta = 1-\sin^2\theta = 1-\frac{9}{25} = \frac{16}{25}$.

34 [Intermediate] • Target: Proving trigonometric identities

Evaluate $\cos(60^\circ + 45^\circ)$ using a sum/difference formula.

a. $2-64$ ✓

b. $\frac{1}{2}$

c. 22

d. $6-24$

$(60^{+}45^{)} = 60^{4}5^{-}60^{4}5^{=}2-64$.

35 [Intermediate] • Target: Solving equations that combine several identities

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$ ✓

c. 1

d. $\frac{\sqrt{2}}{2}$

The equation of a unit circle centered at the origin is $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $y = \frac{\sqrt{3}}{2}$. Substituting this into the equation of the unit circle: $x^2 + (\frac{\sqrt{3}}{2})^2 = 1$ $x^2 + \frac{3}{4} = 1$ $x^2 = 1 - \frac{3}{4}$ $x^2 = \frac{1}{4}$ Since point P lies in the first quadrant, its x-coordinate must be positive. Therefore, $x = \sqrt{\frac{1}{4}} = \frac{1}{2}$. Alternatively, we are given that the angle θ formed by the ray from the or.

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36 [Intermediate] • Target: A well-designed word problem: alternating current

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?

a. $\cos\theta$ ✓

b. $\sin\theta$

c. $\tan\theta$

d. 1

By definition, the coordinates of a point $P(x, y)$ on the unit circle corresponding to an angle θ (measured counterclockwise from the positive x-axis) are given by $(\cos\theta, \sin\theta)$. Therefore, the x-coordinate of the point P is $\cos\theta$.

37 [Intermediate] • Target: Compound angle formulas

In ΔABC, $\angle A = 60^\circ$, $\angle B = 80^\circ$, and side $a = 5$. Find side b (to one decimal place) using the Law of Sines.

a. **5.7** ✓

b. 6.7

c. 7.7

d. 4.7

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{5\sin 80^\circ}{\sin 60^\circ} \approx 5.7$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Double angle formulas • Cross-Lesson: 7.01 Vectors

In triangle ABC, point D lies on side BC such that AD is an angle bisector of angle A. Let $AB = c$, $AC = b$, $BD = m$, and $CD = n$. Prove that $\frac{m}{n} = \frac{c}{b}$ using a combination of the Sine Rule and the properties of angle bisectors.

a. **This statement is a direct consequence of the Sine Rule applied to triangles ABD and ACD.** ✓

b. This statement can be proven by applying the Law of Cosines to triangles ABD and ACD and equating expressions for AD^2.

c. This statement is true only if triangle ABC is isosceles with $b = c$.

d. This statement requires the use of vector methods for its proof.

Let $\angle BAD = \angle CAD = \alpha$. In triangle ABD, by the Sine Rule, we have $\frac{m}{\sin\alpha} = \frac{c}{\sin\angle ADB}$. Therefore, $m = \frac{c\sin\alpha}{\sin\angle ADB}$. In triangle ACD, by the Sine Rule, we have $\frac{n}{\sin\alpha} = \frac{b}{\sin\angle ADC}$. Therefore, $n = \frac{b\sin\alpha}{\sin\angle ADC}$. Since $\angle ADB$ and $\angle ADC$ are supplementary angles, $\sin\angle ADB = \sin(180^\circ - \angle ADC) = \sin\angle ADC$. Now, we can form the ratio.

39 [Challenging] • Target: Solving equations using double angle formulas • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$ ✓

d. $\frac{4}{5}$

On the unit circle, any point P(x, y) satisfies the equation $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $\frac{3}{5}$ and P is in the second quadrant. For a point $P(\cos\theta, \sin\theta)$ on the unit circle, we have $x = \cos\theta$ and $y = \sin\theta$. Thus, we have $\sin\theta = \frac{3}{5}$. Substituting this into the unit circle equation: $\cos^2\theta + (\frac{3}{5})^2 = 1$. $\cos^2\theta + \frac{9}{25} = 1$. $\cos^2\theta = 1 - \frac{9}{25}$. $\cos^2\theta = \frac{16}{25}$..

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40 [Challenging] • Target: The R sin(θ+α) form • Cross-Lesson: 9.01 Complex Numbers

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$ ✓

d. 1

The coordinates of a point on the unit circle corresponding to an angle θ are given by $(\cos\theta, \sin\theta)$. In this case, $\theta = \frac{\pi}{3}$. Therefore, the y-coordinate of point P is $\sin\left(\frac{\pi}{3}\right)$. We know that $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$.

41 [Challenging] • Target: Using the R-form to find extrema and solve equations • Cross-Lesson: 1.01 Functions: Modulus & Inverse

In ΔABC, $a = 8$, $b = 10$, $c = 12$. Find $\angle A$ (to one decimal place) using the Law of Cosines.

a. **41.4°** ✓

b. 46.4°

c. 138.6°

d. 36.4°

By the Law of Cosines: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{100 + 144 - 64}{240} \approx 0.75$, so $A = \cos^{-1}(0.75) \approx 41.4°$.

42 [Challenging] • Target: Proving trigonometric identities • Cross-Lesson: 10.01 Numerical Methods

In ΔABC, $a = 5$, $b = 6$, $c = 7$. Find $\angle A$ (to one decimal place) using the Law of Cosines. (trial 2)

a. 49.4°

b. 135.6°

c. **44.4°** ✓

d. 39.4°

By the Law of Cosines: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{36 + 49 - 25}{84} \approx 0.714$, so $A = \cos^{-1}(0.714) \approx 44.4°$.

43 [Challenging] • Target: Solving equations that combine several identities • Cross-Lesson: 11.01 Differential Equations

In ΔABC, $b = 10$, $c = 7$, and $\angle A = 120°$. Find side a (to one decimal place) using the Law of Cosines.

a. 13.8

b. **14.8** ✓

c. 15.8

d. 12.2

By the Law of Cosines: $a^2 = b^2 + c^2 - 2bc\cos A = 10^2 + 7^2 - 2(10)(7)\cos 120° \approx 219.0$, so $a \approx 14.8$.

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44 [Challenging] • Target: A well-designed word problem: alternating current • Cross-Lesson: 12.01 Poisson Distribution

Use the sum formula to find $\sin(30^\circ + 60^\circ)$.

a. $\frac{11}{10}$

b. $\frac{1}{2} + \frac{\sqrt{3}}{2}$

c. -1

d. **1** ✓

$\sin(30^\circ + 60^\circ) = \sin 90^\circ = 1$.

45 [Challenging] • Target: Compound angle formulas • Cross-Lesson: 13.01 Continuous Random Variables

In triangle ABC, angle ABC = θ degrees, and angle ACB = 2θ degrees. The lengths of sides AB and AC are 10 cm and 15 cm respectively. The length of BC is denoted by x cm. Given that the area of triangle ABC is 60 cm^2, find the value of x, to 3 significant figures.

a. 13.7

b. 11.4

c. **12.8** ✓

d. 14.9

Let the angles of triangle ABC be denoted as follows: $\angle ABC = \theta$, $\angle ACB = 2\theta$. By the sine rule, $\frac{AC}{\sin(\theta)} = \frac{AB}{\sin(2\theta)}$. Substituting the given values, $\frac{15}{\sin(\theta)} = \frac{10}{\sin(2\theta)}$. Using the double angle identity, $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$, we get $\frac{15}{\sin(\theta)} = \frac{10}{2\sin(\theta)\cos(\theta)}$. Since $\sin(\theta) eq 0$ for a non-degenerate triangle, we can cancel $\sin(\theta)$ from both sides.

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